August 6, 2026

Dear Hiring Team,

I am writing to apply for the Geodata Scientist position at ALS Geoanalytics. With a Master of Science in Computer Science from the University of Calgary and applied experience in geospatial machine learning, remote sensing, spatial data processing, technical writing, and scientific software development, I am excited by the opportunity to help ALS turn complex geoscience datasets into practical exploration insights.

My graduate research focused on scalable methods for managing, processing, and analyzing large Earth observation datasets. Working with BigGeo and Mitacs, I built C++ and Python software for satellite imagery processing, automated SAR and optical imagery preprocessing with GDAL, Rasterio, SNAP, and satellite data APIs, and developed DGGS-based encoding pipelines that convert imagery into structured tensor data. This work required careful handling of geospatial data constraints, multiresolution spatial indexing, reproducibility, and performance, which aligns closely with ALS Geoanalytics' focus on geoscience data integration and analytical workflows.

I also bring machine learning and model evaluation experience that is directly relevant to prospectivity analysis and predictive modelling. In my DEM super-resolution project, I trained PyTorch ResNet models, built data collection and preprocessing workflows using Google Earth Engine and Rasterio, and evaluated results with terrain-aware metrics such as RMSE, slope, TRI, TPI, and wavelet-based errors. I also developed a Polyscope-based 3D visualization tool to inspect elevation reconstructions from multiple angles, strengthening my ability to connect model outputs with spatial interpretation.

Beyond the technical fit, I enjoy communicating complex analytical work clearly. My first-author Remote Sensing publication and M.Sc. thesis required turning research methods, results, and limitations into structured writing for scientific audiences. As a teaching assistant, I supported students in big data applications, data-at-scale workflows, and database systems, which helped me explain technical concepts to audiences with different backgrounds. I would bring that same care to client deliverables, technical reports, presentations, and collaboration with multidisciplinary teams.

Thank you for considering my application. I would welcome the opportunity to discuss how my background in geospatial data systems, Python-based scientific computing, machine learning workflows, and research communication can contribute to ALS Geoanalytics' consulting projects and platform development.

Sincerely,
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